SQL FUNDAMENTALS

PRACTICAL PROJECT • ONLINE SHOP PRODUCT DATABASE

PROJECT OBJECTIVE

Demonstrate your understanding of data, tables, rows, columns, databases, SQL, queries, SELECT, FROM, SELECT *, WHERE, and filtering using = and >. **No SQL software is required.**

PART 1 — CREATE YOUR TABLE

Imagine you are helping to build a database for an online shop. Create a table called **products** with **at least 8 products**. Complete the table below using your own product data.

Column	Description
id	Product ID
name	Product name
category	Product category
price	Product price
stock	Quantity available

id	name	category	price	stock
1.	Nike Running Shoes	Footwear	85,000	8
2.	Leather Handbag	Bags	40,000	12
3.	jacket	Clothing	45,000	8
4.	Cotton T-shirt	Clothing	80,000	16
5.	Ankara Dress	Clothing	30,000	14
6.	Face Moisturizer	Skincare	1000	4
7.	Liquid Soap	Household	2000	6
8.	Vitamin-c Serum	Skincare	30,000	13
9.	Looking Good Cream	Skincare	35,000	15
10.	Electric Kettle	Kitchenware	25,000	4

PART 2 — UNDERSTANDING THE TABLE

Task 1 — How many rows and columns are in your table? ✓

There are 10 rows and 5 columns in the Table.

Task 2 — What does a row represent and what does a column represent? ✗

A row represent a single product /record in the table.

PART 3 — BASIC SQL QUERIES

Task 3 — Display the names of all products. ✓

```
SELECT name
FROM products;
```

Task 4 — Display the names, categories, and prices of all products. ✓

```
SELECT name, category, price
FROM products;
```

Task 5 — Display all information from the products table. ✓

```
SELECT *
FROM products;
```

PART 4 — FILTERING WITH WHERE

Task 6 — Show the names of all products in the Phones category. ✓

```
SELECT name
FROM products
WHERE category = 'phones';
```

Task 7 — Show the names of products whose price is greater than 100,000. ✓

```
SELECT name
FROM products
WHERE price > 100000;
```

Task 8 — Show the names and stock quantities of products whose stock is greater than 10. ✓

```
SELECT name, stock
FROM products
WHERE stock > 10;
```

PART 5 — UNDERSTANDING SQL

Task 9 — In your own words, explain what each of the following does: ✓

SELECT	FROM	WHERE
What information I want to see?	Which table to look in?	What condition should it meet?

PART 6 — REAL-WORLD CHALLENGE

Task 10 ✓

The shop manager asks:

"Show me the names of all products in the Laptop category that cost more than 200,000."

Write the SQL query that would answer the question. **You do not need to run it.**

```
SELECT name
FROM products
WHERE category = 'laptop'
AND price > 200000;
```

Explain your query: Briefly explain what each part of your query does.

SELECT name (show only the product names**),** **FROM product (**Get the information from the products table**)**

WHERE category = 'laptop' (choose only products whose category is laptop**), AND price > 200000(**the price must be 200000).

SUBMISSION REQUIREMENTS

- Complete the products table with at least 8 products.
- Answer Tasks 1–10.
- Write all SQL queries on this assessment sheet.
- No SQL software, database application, or screenshot is required.
- Use your own words when explaining concepts.

INSTRUCTOR ASSESSMENT

Area	Marks
Understanding tables, rows and columns	15
Basic SELECT and FROM queries	20
SELECT *	10
WHERE and filtering	25
Understanding SQL concepts	10
Real-world challenge	15
Explanation and presentation	5
TOTAL	**100**

Performance Levels: 90–100 Excellent • 75–89 Good • 60–74 Developing • Below 60 Needs additional practice

SQL ASSESSMENT · GRADED COPY

ZAINAB SQL

TOTAL SCORE: 94/100

PERFORMANCE: Excellent

MARKING BREAKDOWN

Task 1 · Rows/columns	7/7
Task 2 · Row/column meaning	4/8
Task 3 · SELECT name	10/10
Task 4 · SELECT name, category, price	10/10
Task 5 · SELECT *	10/10
Task 6 · WHERE category	8/8
Task 7 · WHERE price >	8/8
Task 8 · WHERE stock >	9/9
Task 9 · SQL concepts	10/10
Task 10 · Real-world query	14/15
Explanation/presentation	4/5

MARKER COMMENTS

Main issue: Task 2 is incomplete (column definition omitted); Task 10 explanation should say price > 200,000, not exactly 200,000.

AI-use comment: There is no reliable way to prove AI use from the PDF alone. If AI was used, it appears to have been used sensibly and then mostly checked; the SQL is strong, though the Task 10 explanation says ·must be 200000· instead of ·greater than 200000.·

Legend: green tick = correct; red cross = incorrect/needs correction.